Loops: for, range and while

Programming · Level 2 · Lesson 3



Today we will

- Repeat lines with for and range.
- Say exactly where range stops.
- Build a total in a loop, and use while.

Remember

- range(1, 5) gives 1, 2, 3, 4: it stops before 5.
- total = 0 before the loop; total = total + n inside it.
- while repeats until its condition is false.

Be the loop

1. Warm-up (5 min): Clap a pattern five times; write it as a loop.
2. Teach (10 min): for, range and a times table.
3. Be the loop (10 min): Pupils act out a running total.
4. Predict and run (10 min): Change the range, predict, run.
5. Wrap-up (5 min): Where does range stop? Exit ticket.

No devices are needed until “Predict and run”: number cards and chalk are enough, and one phone runs the programs.

Let’s discuss

1. Why does range stop before its second number?
2. When would you use while instead of for?
3. Which job in our school repeats, and could be a loop?

Exit ticket

- Write a loop that prints 2, 4, 6, 8, 10, and say which range you used.